

01_SNES_BUS_AND_CIC

Arquivo: 01_SNES_BUS_AND_CIC.kicad_sch

02_LEVEL_SHIFTERS_AND_FLASH

Arquivo: 02_LEVEL_SHIFTERS_AND_FLASH.kicad_sch

03_RP2350_SUBSYSTEM

Arquivo: 03_RP2350_SUBSYSTEM.kicad_sch

04_ESP32C6_WIFI_IOT

Arquivo: 04_ESP32C6_WIFI_IOT.kicad_sch

05_PERIPHERALS_AND_POWER

Arquivo: 05_PERIPHERALS_AND_POWER.kicad_sch

SUPER FAMICOM / SNES SMART HOME INTERFACE CARTRIDGE V1.0

System Overview:

- Sheet 1: 62-Pin SNES Bus Edge Connector (2.50mm pitch, 1.2mm PCB) + SuperCIC PIC12F629
- Sheet 2: 74LVC541 + SN74LVC8T245 Level Shifters + 4Mbit Parallel Boot Flash (SST39VF040)
- Sheet 3: RP2350B MCU Subsystem (Dual Cortex-M33, 520KB SRAM Mailbox, 16MB QSPI, USB-C)
- Sheet 4: ESP32-C6-MINI-1 Wi-Fi 6 / BLE / 802.15.4 Module (Home Assistant REST/WS, MQTT)
- Sheet 5: Peripherals (MicroSD SPI, DS3231MZ RTC, SP3485 RS-485, Qwiic I2C) + SY8089 5V->3.3V 2A Buck

Targeted for PCBWay 4-Layer 1.2mm ENIG 30-deg Bevel
RP2350B + ESP32-C6 + Parallel Flash + Level Shifting

SNES SmartHome Project

Sheet: /
File: snes_smarthome_cartridge.kicad_sch

Title: Super Famicom Smart Home & Diagnostics Cartridge

Size: A3	Date: 2026-09-10	Rev: V1.0
KiCad E.D.A. 10.0.6		Id: 1/6

- ## SNES SmartHome Project

Title: SNES 62-Pin Cartridge Connector & SuperCIC

Size: A3

Date: 2026-09-10

Rev: V1.0

KiCad E.D.A. 10.0.6

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A	SHEET 4: ESP32-C6 WI-FI 6 / BLE / THREAD COPROCESSOR							
B	Module: Espressif ESP32-C6-MINI-1 (or ESP32-C6-WROOM-1)							
	- 32-bit RISC-V single-core @ 160MHz, 512KB SRAM, 4MB/8MB Flash - Wi-Fi 6 (2.4 GHz 802.11ax), Bluetooth 5 (LE), IEEE 802.15.4 (Thread/Zigbee) - Built-in certified PCB antenna (placed at the top edge of cartridge with keepout)							
C	Role in SNES Smart Home:							
	- Connects to local Wi-Fi and establishes Home Assistant session - Handles TLS encryption, JSON / WebSocket / REST parsing, or MQTT - Translates Home Assistant states into compact binary protocol (SH 01 + CRC16) - Pushes snapshot to RP2350 mailbox via high-speed SPI / UART - Receives command packets from RP2350 and dispatches HA service calls							
D	Interface to RP2350:							
	- SPI: ESP_MOSI (GPIO19), ESP_MISO (GPIO20), ESP_SCK (GPIO21), ESP_CS (GPIO18) - UART: ESP_TX (GPIO16) -> RP2350_RX, ESP_RX (GPIO17) <- RP2350_TX - Handshake IRQ: ESP_INT (GPIO22) to alert RP2350 of new states							
E	Power & Decoupling:							
	- 3.3V with 10uF ceramic + 100nF ceramic + bulk 47uF tantalum nearby - CHIP_PU with 10k pullup and 1uF delay capacitor for clean power-on reset - Tactile buttons: BOOT (GPIO9) and RESET (CHIP_PU) for firmware flashing - Onboard Status Indicator: WS2812B / SK6812MINI addressable RGB LED							
F	1	2	3	4	5	6	7	8

SNES SmartHome Project		
Sheet: /04_ESP32C6_WIFI_IOT/		
File: 04_ESP32C6_WIFI_IOT.kicad_sch		
Title: ESP32-C6 Wi-Fi 6 / IoT Coprocessor		
Size: A3	Date: 2026-09-10	Rev: V1.0
KiCad E.D.A. 10.0.6	Id: 3/6	

	1	2	3	4	5	6	7	8
A	SHEET 5: PERIPHERALS (MICROSD, RTC, RS-485, I2C) & 5V->3.3V BUCK							
B	1. Power Supply Subsystem (High Efficiency Synchronous Buck): - Input: +5V_SNES from cartridge edge (pins 27, 58) - Reverse polarity / overcurrent protection: SS34 Schottky diode / PTC resettable fuse (1.5A) - U9: Silergy SY8089AAAC (or AP62200 / TPS62088) Synchronous Step-Down DC-DC Converter - Package: SOT-23-5, 1.5MHz switching frequency, 94% efficiency, 2.0A continuous output - Inductor: L1 = 2.2uH SMD high-current power inductor (CDRH3D16 or 0805) - Feedback: R1 = 100k, R2 = 22k for calibrated 3.30V VCC rail - Input filter: 10uF 0805 X7R + 47uF low-ESR tantalum - Output filter: 22uF 0805 X7R + 47uF low-ESR tantalum - Much cooler than linear regulators (AMS1117), preserves SNES internal power brick!							
C	2. MicroSD Card Socket: - J4: Push-Pull MicroSD connector, connected via SPI/SDIO to RP2350 - Enables logging, configuration files (wifi.txt / ha_config.json), and sound/ROM assets - 10k pullups on CS and CMD lines, dedicated 100nF + 10uF decoupling							
D	3. High-Precision RTC (Real-Time Clock): - U10: Maxim DS3231MZ+ (SOIC-8) with integrated MEMS resonator, +-5ppm accuracy - I2C connected to RP2350 (SCL, SDA with 4.7k pullups to 3.3V) - BT1: CR1220 3V coin cell battery clip for timekeeping when console is powered off							
E	4. RS-485 Industrial / Home Automation Bus: - U11: SP3485CN-L (SOIC-8, 3.3V half-duplex RS-485 transceiver) - Connected to RP2350 UART + direction control pin - J5: 3-pin 3.5mm screw terminal block (A, B, GND) with 120 ohm termination jumper - SM712 bidirectional TVS diode array for lightning & ESD protection on RS-485 differential lines							
F	5. External Sensor / Display Expansion: - J6: Qwiic / STEMMA QT 4-pin 1.0mm JST-SH connector (GND, 3.3V, SDA, SCL) - Plug-and-play connection for external BME280, OLED, or radar sensors							
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SNES SmartHome Project

Sheet: /05_PERIPHERALS_AND_POWER/
File: 05_PERIPHERALS_AND_POWER.kicad_sch

Title: **Peripherals (MicroSD, RTC, RS-485, I2C) & Power**

Size: A3

Date: 2026-09-10

Rev: **V1.0**

KiCad E.D.A. 10.0.6

Id: 5/6

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A	SHEET 3: RP2350B DUAL-CORE MCU SUBSYSTEM							
B	MCU Selection: Raspberry Pi RP2350B (QFN-80)							
	- 48 GPIOs allow dedicated bus taps and rich peripheral connectivity - Dual ARM Cortex-M33 cores @ 150MHz (overclockable to 250MHz+) - 520KB on-chip SRAM: perfectly hosts the Home Assistant Mailbox & dynamic snapshot - 12 PIO State Machines for real-time bus interception, cycle-accurate timing							
C	Pin Allocation:							
	- GPIO0..GPIO7: BUS_D0..BUS_D7 (Bidirectional data bus for SRAM/mailbox access) - GPIO8..GPIO23: BUS_A0..BUS_A15 (Address bus snooping & decoding) - GPIO24: BUS_/RD, GPIO25: BUS_/WR, GPIO26: BUS_/ROMSEL, GPIO27: BUS_PHI2 - GPIO28: BUS_/IRQ (Open-drain to SNES /IRQ line for instant event notification) - GPIO29..GPIO32: SPI0 to ESP32-C6 (SCK, MOSI, MISO, CS) - GPIO33, GPIO34: UART0 to ESP32-C6 (TX, RX) - GPIO35, GPIO36: I2C0 to RTC & Qwiic / STEMMA QT (SDA, SCL) - GPIO37..GPIO40: MicroSD (SPI1: SCK, MOSI, MISO, CS) - GPIO41, GPIO42: RS-485 (TX, RX, DE/RE direction) - GPIO43..GPIO47: Status LEDs, Jumper inputs							
D	QSPI Flash Memory:							
	- U7: Winbond W25Q128JVSQ (16MB / 128Mbit SPI/QSPI NOR Flash, SOIC-8 208mil) - Connected to RP2350 QSPI pins with 10k pullups on CSn							
E	Oscillator & Power:							
	- Y1: 12.000 MHz SMD 3225 Crystal + 2x 15pF COG capacitors - Core DVDD (1.1V): Internal regulator with 1uF + 100nF decoupling - IOVDD (3.3V): Decoupled with 100nF per VDD pin + 10uF bulk							
F	USB-C Programming & Diagnostics:							
	- J1: USB4110 / Type-C 16-pin connector with 5.1k pulldowns on CC1/CC2 - USBLC6-2SC6 ESD protection on D+/D- + 27 ohm series resistors - BOOTSEL & RESET tactile buttons for UF2 drag-and-drop firmware upgrades							
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SNES SmartHome Project		
Sheet: /03_RP2350_SUBSYSTEM/		
File: 03_RP2350_SUBSYSTEM.kicad_sch		
Title: RP2350B MCU Subsystem & Mailbox		
Size: A3	Date: 2026-09-10	Rev: V1.0
KiCad E.D.A. 10.0.6	Id: 6/6	